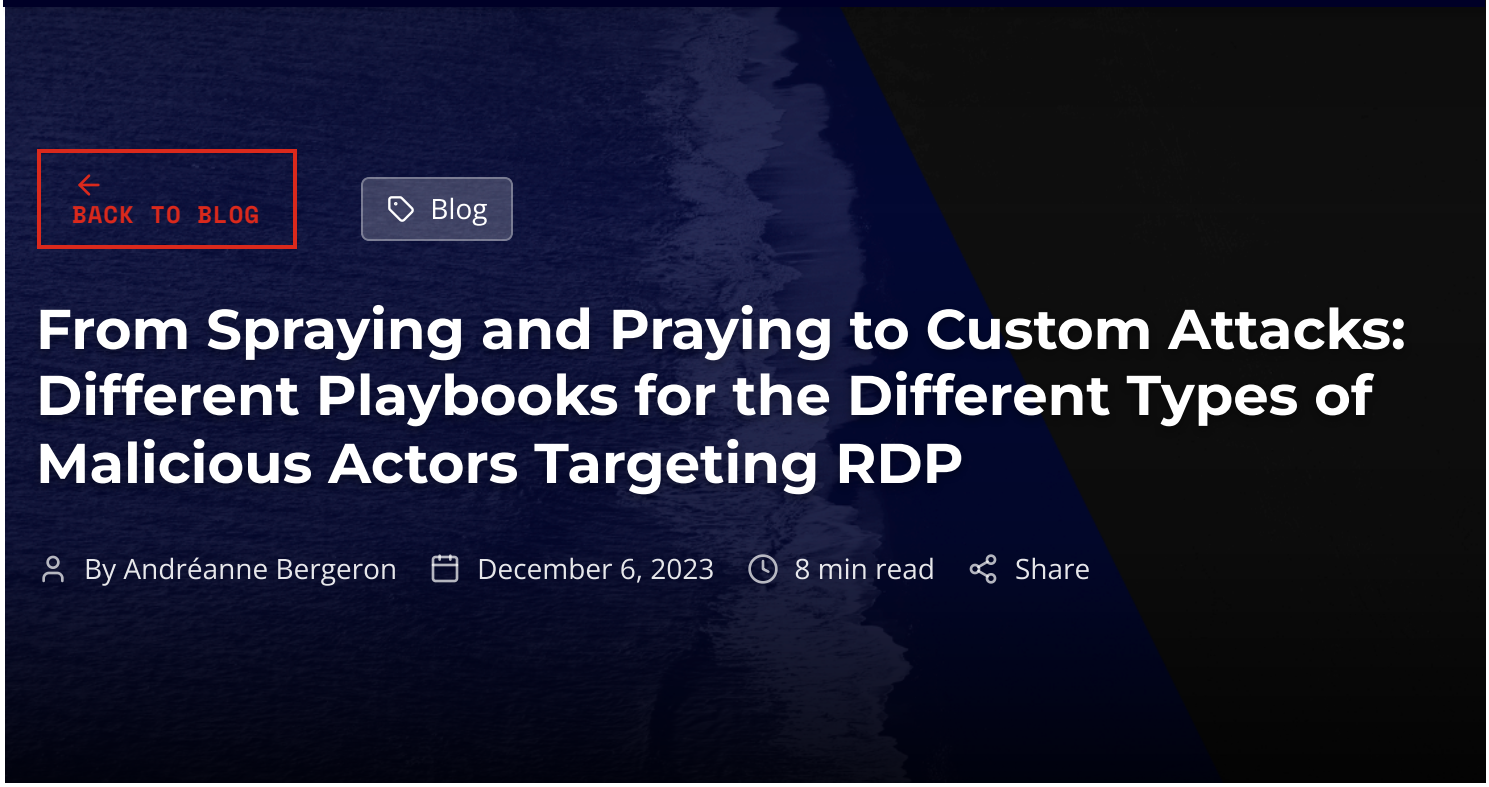


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Blog

From Spraying and Praying to Custom Attacks: Different Playbooks for the Different Types of Malicious Actors Targeting RDP

By Andréanne Bergeron | December 6, 2023 | 8 min read | Share

Characterizing attackers gets us closer to reveal who they are. Our study categorizes the behavior of Remote Desktop Protocol (RDP) attackers. Based on 3.4 million login attempts, we reveal five different clusters of brute-force attacks strategies.

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elements observed in the sample in terms of strategies. Each element used to form the profiles is presented in the table below.

Variable	**Definition**
Identical username and password	The attacker is using the same character sequence for the username and the password.
Attacks are customized to the target	There is the presence of the honeypot certificate name or computer name in the credential used.
Delay between attacks	There is a short delay between attempt logins.
High number of attacks	More than 50 login attempts by the same attacker.
High proportion in RockYou2021	More than 60% of the passwords used in attempt logins are present in RockYou 2021.
Username-password combination reused	The attackers use the same username-password combination more than once.

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opposite, but rather that it is strongly not associated with the cluster. A name and an image have been attributed to each cluster to describe them better.

Cluster 1: The delayed timer



The element that characterizes this cluster is the fact that the attackers impose a delay between attacks. Usually, automated attacks include many attempted logins per second. However, some systems block, or slow down successive login attempts that are too quick for human behavior. Imposing a small delay between login attempts might be a great strategy to avoid these countermeasures and therefore avoid detection. This allows them to keep trying to brute-force their way in for a longer time.

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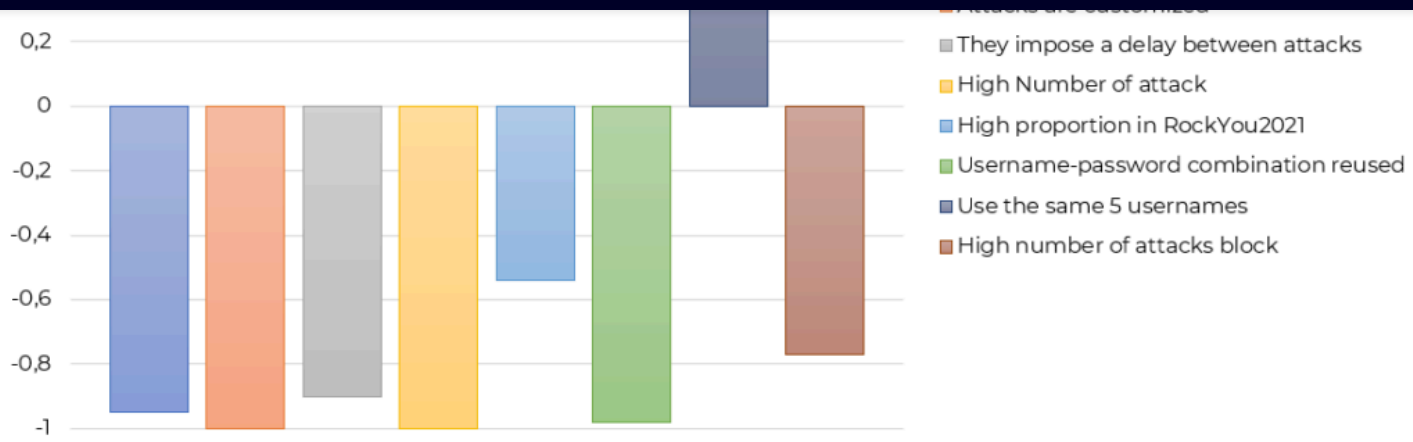
Cluster 2: Vertical sprayer



Cluster 2 is characterized by the fact that they use the same 5 usernames with a bunch of

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When it comes to the art of painting with spray cans, horizontal spraying is known to give better results. The reason is that it is easier to keep your wrist from sweeping if you are spraying horizontally rather than vertically. However, when it comes to brute-forcing systems, which technique is the most efficient is not so clear. On our systems, attackers had limited prior information on their target so we can assume that they strongly relied on common usernames in the short list they tried.

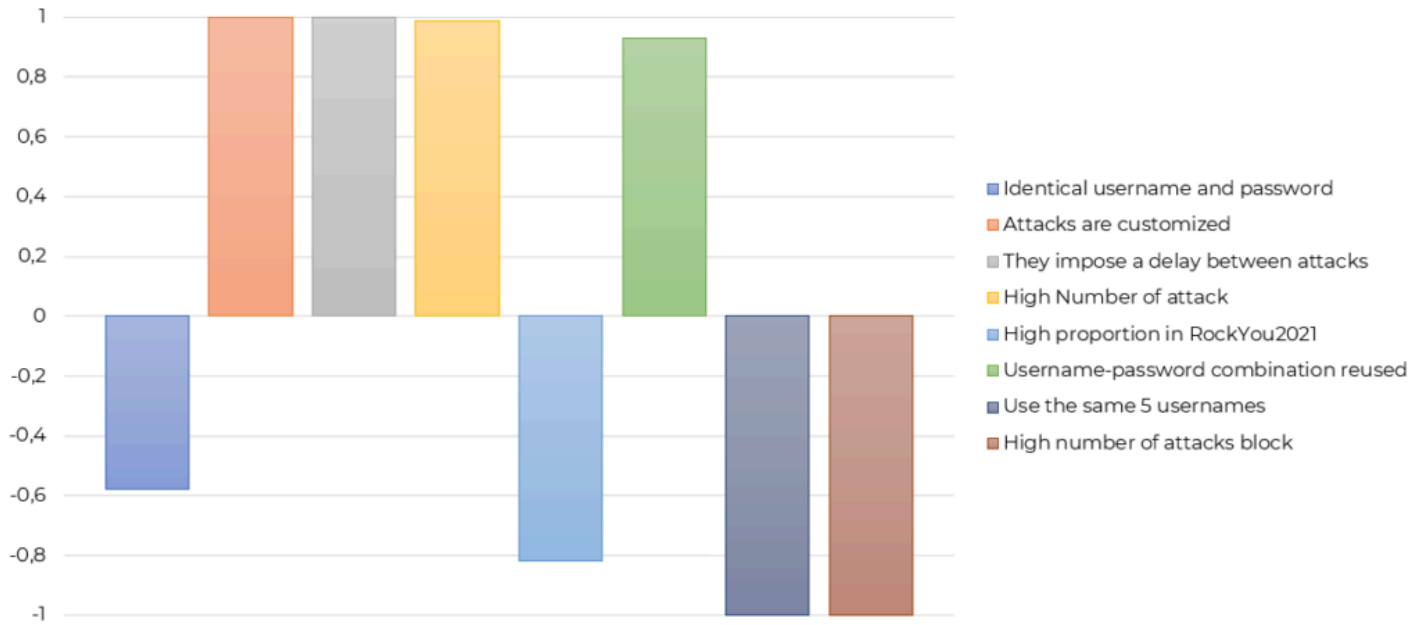
Cluster 3: Targeting crew

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sophisticated compared to the other clusters. They went to find the information and try to leverage it to access the system. They also might use the same character chain for the username and the password, assuming that the RDP administrator uses the same for both.

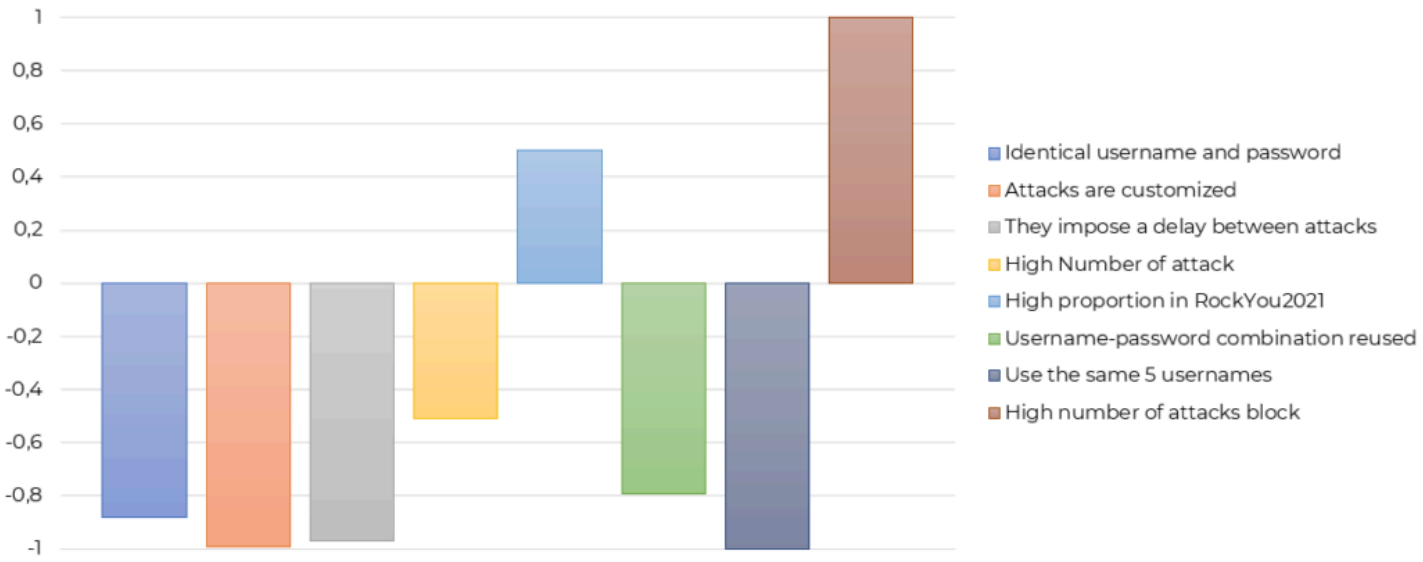


Cluster 4: Basic automation

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tend to use data leaks that are readily available on the internet in their login attempts, and they need to intervene to launch their list instead of them to be automated.

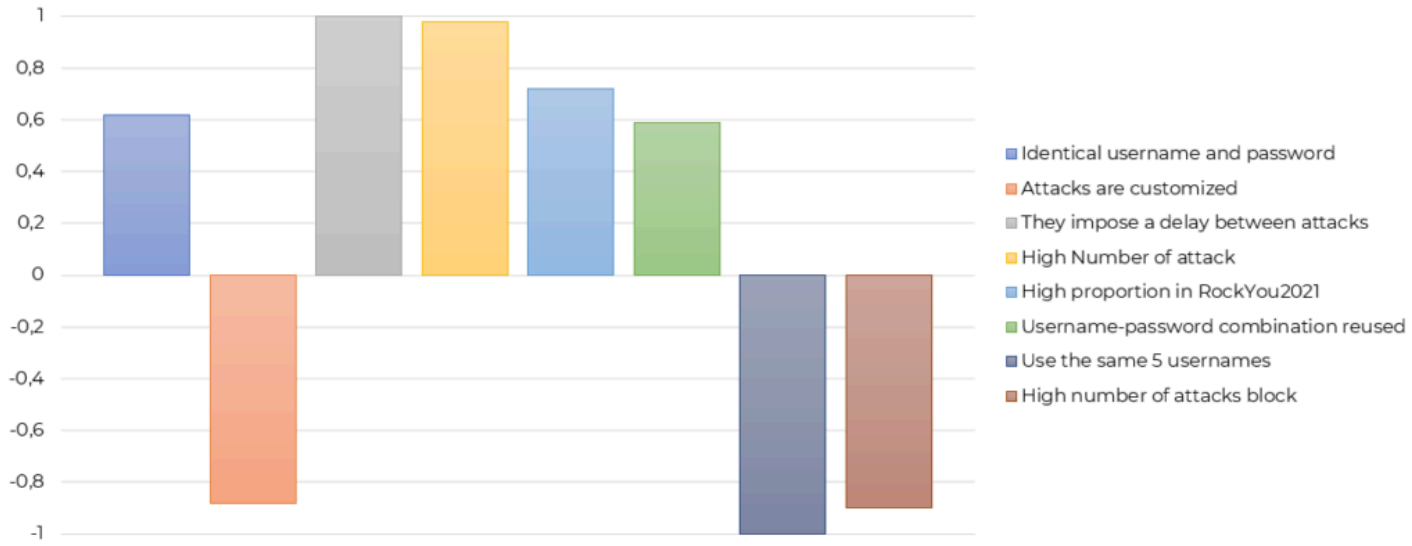


Cluster 5: Lost in the loop

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once against our system. Since they are heavy attackers, it gives the impression that they might be a team of hackers working on the same targets and not communicating well on what has been tried. Or it can be a disorganized and not so sophisticated hacker. In any case, they look like the confused ones.



Mitigation related to clusters

In cluster 1, attackers impose a delay to imitate human behavior and avoid detection. However, the delay between login attempts is short so the attacks are still efficient. The more attempts

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One of the most efficient countermeasures for blocking heavy attackers' brute forcing is a lockout policy like the one established by Windows 11 by default. That policy automatically locks user accounts after 10 failed sign-in attempts. The account will be blocked for 10 minutes which will decrease the efficiency of brute forcing significantly and will be much more time consuming for attackers even if they use automation to try again after 10 minutes. Imposing a delay of a few seconds will no longer be efficient.

Conclusion

Our system is targeted by a login attempt on average every two seconds. The overall strategy of attack is, however, highly different from one threat actor to another. Documenting the different types of attackers is our way of sharing information to start a discussion about different prevention solutions. If you have mitigation ideas, do not hesitate to share!

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